FOREST FIRE RISK MANAGEMENT

FIRE TRIANGLE AND CONTROL → FIRE-TYPE MATRIX → FIRE-BEHAVIOUR TRIANGLE → IGNITION GOVERNANCE MAP → FSI ALERT RAIL → INITIAL RESPONSE CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 — FIRE TRIANGLE AND CONTROL

FIRE TRIANGLE AND CONTROL · FUEL + HEAT + OXYGEN · REDUCE UNSAFE IGNITION · FUEL CONTINUITY · INTERRUPT HEAT TRANSFER WHERE SAFE · FIRE-FREE FOREST IS NOT A UNIVERSAL GOAL

FUEL + HEAT + OXYGEN → COMBUSTION

PREVENT → reduce unsafe ignition / fuel continuity → RESPOND → interrupt heat transfer where safe → LIMIT → FIRE-FREE FOREST IS NOT A UNIVERSAL GOAL

HAZARD / RISK / RESILIENCE CONCEPT
- FUEL + HEAT + OXYGEN → COMBUSTION
- PREVENT → reduce unsafe ignition / fuel continuity

INSTITUTIONAL OR COMMUNITY RESPONSE
- RESPOND → interrupt heat transfer where safe
- LIMIT → FIRE-FREE FOREST IS NOT A UNIVERSAL GOAL

CAPACITY / STATUS / EVIDENCE LIMIT
- FUEL + HEAT + OXYGEN → COMBUSTION
- PREVENT → reduce unsafe ignition / fuel continuity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → FIRE-FREE FOREST IS NOT A UNIVERSAL GOAL.

01

STAGE 01 — FIRE-TYPE MATRIX

FIRE-TYPE MATRIX · LITTER GRASS SHRUB LAYER · ORGANIC MATTER BELOW · WITHIN SURFACE · TREE CANOPY · VERTICAL FUEL AND CONDITIONS CONNECT LAYERS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
SURFACE → litter grass shrub layer	CROWN → tree canopy	SURFACE → litter grass shrub layer
GROUND → organic matter below / within surface	TRANSITION → vertical fuel and conditions connect layers	GROUND → organic matter below / within surface

HAZARD / ACTOR / INSTRUMENT
- SURFACE → litter grass shrub layer
- GROUND → organic matter below / within surface

DECISIVE RISK OR MANDATE DISTINCTION
- CROWN → tree canopy
- TRANSITION → vertical fuel and conditions connect layers

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- SURFACE → litter grass shrub layer
- GROUND → organic matter below / within surface

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSITION → vertical fuel and conditions connect layers.

02

STAGE 02 — FIRE-BEHAVIOUR TRIANGLE

FIRE-BEHAVIOUR TRIANGLE · DRYNESS WIND HUMIDITY TEMPERATURE · TYPE LOAD MOISTURE CONTINUITY · SLOPE ASPECT VALLEY ACCESS · SPREAD INTENSITY SMOKE AND RESPONSE LIMITS

WEATHER → dryness wind humidity temperature

FUEL → type load moisture continuity → TOPOGRAPHY → slope aspect valley access → INTERACTION → spread intensity smoke and response limits

RISK CONDITION / INPUT
- WEATHER → dryness wind humidity temperature
- FUEL → type load moisture continuity

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- TOPOGRAPHY → slope aspect valley access
- INTERACTION → spread intensity smoke and response limits

LAST-MILE / INCLUSION / OUTCOME LIMIT
- WEATHER → dryness wind humidity temperature
- FUEL → type load moisture continuity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTERACTION → spread intensity smoke and response limits.

03

STAGE 03 — IGNITION GOVERNANCE MAP

IGNITION GOVERNANCE MAP · LIGHTNING AND SOURCE-BOUNDED CAUSES · DELIBERATE HUMAN · RESOURCE-USE MOTIVE · ACCIDENTAL HUMAN · ESCAPED HEAT · PREVENTION BY CAUSE NOT GENERIC BLAME

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
NATURAL → lightning and source-bounded causes	ACCIDENTAL HUMAN → escaped heat / equipment / smoking	NATURAL → lightning and source-bounded causes
DELIBERATE HUMAN → land / resource-use motive	RESPONSE → prevention by cause not generic blame	DELIBERATE HUMAN → land / resource-use motive

HAZARD / ACTOR / INSTRUMENT
- NATURAL → lightning and source-bounded causes
- DELIBERATE HUMAN → land / resource-use motive

DECISIVE RISK OR MANDATE DISTINCTION
- ACCIDENTAL HUMAN → escaped heat / equipment / smoking
- RESPONSE → prevention by cause not generic blame

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- NATURAL → lightning and source-bounded causes
- DELIBERATE HUMAN → land / resource-use motive

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESPONSE → prevention by cause not generic blame.

04

STAGE 04 — FSI ALERT RAIL

FSI ALERT RAIL · 1 MODIS · SNPP-VIIRS DETECTION · 2 AUTOMATED PROCESSING AND FOREST FILTER · 3 SMS · MAP DISSEMINATION · 4 FIELD VERIFY

HAZARD / RISK / RESILIENCE CONCEPT
- 1 MODIS / SNPP-VIIRS DETECTION
- 2 AUTOMATED PROCESSING AND FOREST FILTER

INSTITUTIONAL OR COMMUNITY RESPONSE
- 3 SMS / EMAIL / MAP DISSEMINATION
- 4 FIELD VERIFY → COMMAND → RESPONSE

CAPACITY / STATUS / EVIDENCE LIMIT
- 1 MODIS / SNPP-VIIRS DETECTION
- 2 AUTOMATED PROCESSING AND FOREST FILTER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 SMS / EMAIL / MAP DISSEMINATION; 4 FIELD VERIFY → COMMAND → RESPONSE.

05

STAGE 05 — INITIAL RESPONSE CHAIN

INITIAL RESPONSE CHAIN · VERIFY LOCATION AND ACCESS · LOCAL COMMAND + TRAINED PROTECTED CREW · TOOLS WATER COMMUNICATION AND SAFETY · ESCALATE IF CAPACITY · CONDITIONS EXCEEDED

ALERT → VERIFY LOCATION AND ACCESS → LOCAL COMMAND + TRAINED PROTECTED CREW → TOOLS WATER COMMUNICATION AND SAFETY → ESCALATE IF CAPACITY / CONDITIONS EXCEEDED

RISK CONDITION / INPUT
- ALERT → VERIFY LOCATION AND ACCESS
- LOCAL COMMAND + TRAINED PROTECTED CREW

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- TOOLS WATER COMMUNICATION AND SAFETY
- ESCALATE IF CAPACITY / CONDITIONS EXCEEDED

LAST-MILE / INCLUSION / OUTCOME LIMIT
- ALERT → VERIFY LOCATION AND ACCESS
- LOCAL COMMAND + TRAINED PROTECTED CREW

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOOLS WATER COMMUNICATION AND SAFETY; ESCALATE IF CAPACITY / CONDITIONS EXCEEDED.

06

STAGE 06 — COMMUNITY PARTNERSHIP LADDER

COMMUNITY PARTNERSHIP LADDER · CONSULT ON RESOURCE-USE AND IGNITION DRIVERS · TRAIN + EQUIP + COMPENSATE · LINK EARLY REPORTING TO FOREST DEPARTMENT · DO NOT OFFLOAD STATE DUTY

CONSULT ON RESOURCE-USE AND IGNITION DRIVERS

TRAIN + EQUIP + COMPENSATE → LINK EARLY REPORTING TO FOREST DEPARTMENT → DO NOT OFFLOAD STATE DUTY

RISK CONDITION / INPUT
- CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
- TRAIN + EQUIP + COMPENSATE

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- LINK EARLY REPORTING TO FOREST DEPARTMENT
- DO NOT OFFLOAD STATE DUTY

LAST-MILE / INCLUSION / OUTCOME LIMIT
- CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
- TRAIN + EQUIP + COMPENSATE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK EARLY REPORTING TO FOREST DEPARTMENT; DO NOT OFFLOAD STATE DUTY.

07

STAGE 07 — FUEL-BREAK BOUNDARY

FUEL-BREAK BOUNDARY · FIRE LINE · MAINTAINED BREAK IN FUEL CONTINUITY · PRESCRIBED BURN · AUTHORISED PLANNED FUEL TREATMENT · OBJECTIVES CONDITIONS SUPERVISION CONTINGENCY · GUARANTEED CONTAINMENT OR LICENCE FOR IGNITION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
FIRE LINE → maintained break in fuel continuity	BOTH → objectives conditions supervision contingency	FIRE LINE → maintained break in fuel continuity
PRESCRIBED BURN → authorised planned fuel treatment	NEITHER → guaranteed containment or licence for ignition	PRESCRIBED BURN → authorised planned fuel treatment

HAZARD / ACTOR / INSTRUMENT
- FIRE LINE → maintained break in fuel continuity
- PRESCRIBED BURN → authorised planned fuel treatment

DECISIVE RISK OR MANDATE DISTINCTION
- BOTH → objectives conditions supervision contingency
- NEITHER → guaranteed containment or licence for ignition

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- FIRE LINE → maintained break in fuel continuity
- PRESCRIBED BURN → authorised planned fuel treatment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NEITHER → guaranteed containment or licence for ignition.

08

STAGE 08 — ECOLOGY-SUPPRESSION BALANCE

ECOLOGY-SUPPRESSION BALANCE · WHAT ECOSYSTEM · FIRE REGIME · LIFE OR ASSET THREAT · SAFETY-LED RESPONSE · ECOLOGY OBJECTIVE · BOUNDED MANAGEMENT · UNIVERSAL EXCLUSION OR UNIVERSAL BURNING

WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?

LIFE OR ASSET THREAT → SAFETY-LED RESPONSE → FUEL / ECOLOGY OBJECTIVE → BOUNDED MANAGEMENT → TRAP → UNIVERSAL EXCLUSION OR UNIVERSAL BURNING

HAZARD / RISK / RESILIENCE CONCEPT
- WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?
- LIFE OR ASSET THREAT → SAFETY-LED RESPONSE

INSTITUTIONAL OR COMMUNITY RESPONSE
- FUEL / ECOLOGY OBJECTIVE → BOUNDED MANAGEMENT
- TRAP → UNIVERSAL EXCLUSION OR UNIVERSAL BURNING

CAPACITY / STATUS / EVIDENCE LIMIT
- WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?
- LIFE OR ASSET THREAT → SAFETY-LED RESPONSE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?.

09

STAGE 09 — SMOKE AND INTERFACE MAP

SMOKE AND INTERFACE MAP · FOREST EDGE · HOME ROAD POWER TOURISM SITE · HEALTH MESSAGE + EVACUATION + UTILITY ACTION · RESPONDER AND VULNERABLE-GROUP PROTECTION

FIRE → SMOKE / VISIBILITY / TRANSPORT

FOREST EDGE → HOME ROAD POWER TOURISM SITE → HEALTH MESSAGE + EVACUATION + UTILITY ACTION → RESPONDER AND VULNERABLE-GROUP PROTECTION

HAZARD / RISK / RESILIENCE CONCEPT
- FIRE → SMOKE / VISIBILITY / TRANSPORT
- FOREST EDGE → HOME ROAD POWER TOURISM SITE

INSTITUTIONAL OR COMMUNITY RESPONSE
- HEALTH MESSAGE + EVACUATION + UTILITY ACTION
- RESPONDER AND VULNERABLE-GROUP PROTECTION

CAPACITY / STATUS / EVIDENCE LIMIT
- FIRE → SMOKE / VISIBILITY / TRANSPORT
- FOREST EDGE → HOME ROAD POWER TOURISM SITE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOREST EDGE → HOME ROAD POWER TOURISM SITE.

10

STAGE 10 — RESTORATION SEQUENCE

RESTORATION SEQUENCE · SAFETY + BURNT AREA · SEVERITY ASSESSMENT · EROSION RUNOFF AND INVASIVE-SPECIES CONTROL · LIVELIHOOD SUPPORT + ECOLOGICALLY SUITABLE RESTORATION · MONITOR RECOVERY · DO NOT EQUATE WITH PLANTING COUNT

SAFETY + BURNT AREA / SEVERITY ASSESSMENT → EROSION RUNOFF AND INVASIVE-SPECIES CONTROL → LIVELIHOOD SUPPORT + ECOLOGICALLY SUITABLE RESTORATION → MONITOR RECOVERY → DO NOT EQUATE WITH PLANTING COUNT

RISK CONDITION / INPUT
- SAFETY + BURNT AREA / SEVERITY ASSESSMENT
- EROSION RUNOFF AND INVASIVE-SPECIES CONTROL

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- LIVELIHOOD SUPPORT + ECOLOGICALLY SUITABLE RESTORATION
- MONITOR RECOVERY → DO NOT EQUATE WITH PLANTING COUNT

LAST-MILE / INCLUSION / OUTCOME LIMIT
- SAFETY + BURNT AREA / SEVERITY ASSESSMENT
- EROSION RUNOFF AND INVASIVE-SPECIES CONTROL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MONITOR RECOVERY → DO NOT EQUATE WITH PLANTING COUNT.

11

STAGE 11 — FOREST-FIRE ANSWER SPINE

FOREST-FIRE ANSWER SPINE · CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY · ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE · SEPARATE ALERT CAPACITY CONTAINMENT AND RECOVERY OUTCOMES

CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY → PREVENT → DETECT → VERIFY → RESPOND → RESTORE → ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE → SEPARATE ALERT CAPACITY CONTAINMENT AND RECOVERY OUTCOMES

HAZARD / RISK / RESILIENCE CONCEPT
- CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY
- PREVENT → DETECT → VERIFY → RESPOND → RESTORE

INSTITUTIONAL OR COMMUNITY RESPONSE
- ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE
- SEPARATE ALERT CAPACITY CONTAINMENT AND RECOVERY OUTCOMES

CAPACITY / STATUS / EVIDENCE LIMIT
- CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY
- PREVENT → DETECT → VERIFY → RESPOND → RESTORE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE.

E

EXTRA — SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · NATIONAL MASTER PLAN FOR FOREST FIRE CONTROL'S FOUR-PART DESIGN · MISSING DEDICATED INSTITUTION · VISIONIAS EXPLICITLY RECOMMENDS · THE FAST 3.0 INSTITUTIONAL CHAIN AS A GENUINE TECHNOLOGY-TRANSFER · THE 44%-DETECTIONS-IN-3%-OF-LAND-AREA CONCENTRATION (PDF P. 47) · VISIONIAS NAMES SEVERAL SPECIFIC RESEARCH/DATA GAPS · VISIONIAS'S OWN CRITIQUE (SECTION 2) IS FUNDAMENTALLY AN EQUITY/

OPTIONAL RISK / RESILIENCE DEPTH
- National Master Plan for Forest Fire Control's four-part design :
- CAUTION: Missing dedicated institution : VisionIAS explicitly recommends
- The FAST 3.0 institutional chain as a genuine technology-transfer
- The 44%-detections-in-3%-of-land-area concentration (PDF p. 47)

MANDATE, WARNING OR OUTCOME LIMIT
- CAUTION: VisionIAS names several specific research/data gaps: "inadequate
- VisionIAS's own critique (Section 2) is fundamentally an equity/
- CAUTION: Fire-suppression cost versus long-term forest-management
- CAUTION: Wildland-urban interface risk (a concept VisionIAS's own

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- NOT: The "FAST 3.0 exists, therefore forest-fire risk is managed"
- CAUTION: This is a long, source-explicit list of named institutional gaps —
- CAUTION: VisionIAS explicitly names "funding" as one of the "missing"
- Burnt Scar Assessment's semi-automated methodology (under FSI

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels